Guidance n°1: national road vehicle fleet and registration for a bottom-up inventory

Introduction

Deriving the real vehicle fleet for a country is one of the first steps to calculate a bottom-up emission inventory as recommended by the Enhanced Transparency Framework of the UNFCCC.

Most countries have in the first instance often difficulties to derive their real vehicle fleet. This lies into the fact that the real fleet is based on registrations (vehicles added to the fleet) and scrappage (vehicles removed from the fleet). Data is often not properly registered by country authorities, especially scrappage data. There are low incentives for people to deregister their vehicle. Further issues include cars being imported/exported for tourism purposes either from nationals living part of the time abroad or for short/middle-stays length or non-declared car use.

We will first focus on how to compare fleet and new registrations within a country, which is result of the sum of the new registration and scrappage in the past (enabling to create a vehicle fleet model). In the next step, we will elaborate on possible workarounds to eliminate uncertainties linked to scrappage data, and vehicles not in use in the country.

Setting the framework

Stock data are often gathered by the federal road agencies, thanks to obligatory vehicle registration processes in most countries. Data need to be consolidated and gathered din a databank to be further processed as described below.

The time period

Usually, the time period to cover is one year, from 01.01. to 31.12., i.e. all vehicles registered within one specific year. Alternatively, inventories cover one year starting from the middle of one year to the middle of the following year. Both approaches are possible.

How to classify vehicle categories

Vehicles must be classified to allow comparisons and to correlate with mileage and fuel economy data in the context of a bottom-up inventory. The EU provides a detailed and harmonised classification of vehicles based on the UNECE standards (categories named M1, M2, M3 etc.). This classification forms a useful basis for harmonisation, but may be expanded in some countries where vehicles do not clearly fit within the proposed descriptions.

Four hints are provided below to help identify issues with vehicle stock and/or new registrations, as well as possible alternative source as workarounds.